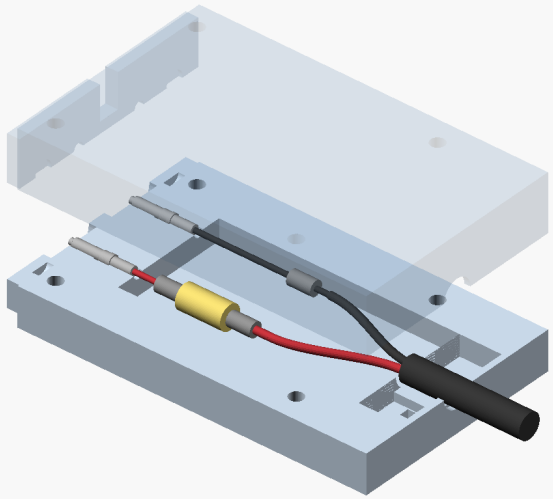
CT-DOCK-001-A | External charging dock / r13

GEOMETRY PROTOTYPE — holding force, assembly process, powered behavior and IP68 remain open

Split housing and sourced component envelopes



Dock: 60 x 36 x 16mm, excluding cable / screw heads.

Two Mill-Max 0962 contacts; 26 AWG pigtails; 0.75A fuse.

Jacket tie bay is separate from the electrical terminations.

Mating and polarity checks

Contact travel	0.239–1.108mm compressed
Bottoming margin	0.416mm minimum
Operating condition	Faces held within 0.05mm gap
Early touch bound	3.250mm face separation
Revised collar key	4.0mm projection; Z=7.7–13.2
Reverse block margin	0.55mm in aligned 180° screen
Wire splay radius	≥10.370mm continuous bound
Wearable main shell	57.4 x 49.4 x 21.0mm, unchanged
Shell including key	61.4 x 49.4 x 21.0mm
Unchanged boot	60.4 x 52.4 x 25.8mm

USB red → F1 → Y=-6 → J3.1 /VIN_5V

USB black → Y=+6 → J3.2 /GND

White/green/drain individually insulated; no USB data.

PCB r05l and both actual 0 Danger / 0 Warning JLC reports are unchanged. This drawing is not a PCB or device release.

The old short/high key fails the reverse-contact checks. Skew, damage and physical misuse are not qualified by this aligned model.

The two springs can require 1.569N at midstroke. Existing magnets have no demonstrated holding force across the plastic walls.

Dry indoor charging only. Crimp tooling, sleeve/joins, cable pull, source/fault behavior and the internal J3 route remain unfinished.

Battery supplier approval, firmware and assembled powered/RF/water/dust tests are still required. No order or physical test was made.